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Safety Data Sheet

1. Chemical product and company identification

Product name : Sodium borohydride

Company information

Name of manufacturer : KANTO CHEMICAL CO., INC.
Address : 2-1, Nihonbashi, Muromachi 2-Chome, Chuo-Ku, Tokyo, 103-0022, JP
Name of section : Reagent Division, Business Administration Department
Telephone number : +81-3-6214-1090
Facsimile number : +81-3-3241-1047
Mail address : BC32@kanto.co.jp
Reference No : 37828

2. Hazards identification

GHS classification

Physical hazards	Substances and mixtures which in contact with water emit flammable gases	Category 1
Health hazards	Acute toxicity (oral)	Category 3
	Skin corrosion/irritation	Category 1A
	Serious eye damage/eye irritation	Category 1
	Specific target organ toxicity (single exposure)	Category 3 (respiratory tract irritation)

Hazard pictograms



Signal word : Danger

Hazard statements : In contact with water releases flammable gases which may ignite spontaneously.
Toxic if swallowed
Causes severe skin burns and eye damage
May cause respiratory irritation

Precautionary statements

Prevention : Keep away from any possible contact with water, because of violent reaction and possible flash fire.
Handle and store contents under inert gas. Protect from moisture.
Do not breathe dust.
Wash hands, forearms and face thoroughly after handling.
Do not eat, drink or smoke when using this product.
Use only outdoors or in a well-ventilated area.
Wear protective gloves/protective clothing/eye protection/face protection.

Response : IF SWALLOWED: Immediately call a POISON CENTER or doctor.
IF SWALLOWED: Rinse mouth. Do not induce vomiting.
IF ON SKIN: Brush off loose particles from skin. Immerse in cool

water .
IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water .
IF INHALED: Remove person to fresh air and keep comfortable for breathing
IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.
Immediately call a POISON CENTER or doctor.
Call a POISON CENTER or doctor if you feel unwell.
Rinse mouth.

Storage : Store in a dry place. Store in a closed container.
Store in a well-ventilated place. Keep container tightly closed.
Store locked up.

Disposal : Dispose of contents/container to hazardous or special waste collection point, in accordance with local, regional, national and/or international regulation.

3. Composition/information on ingredients

Distinction of substance or mixture : Substance

Synonyms : SBH, Sodium tetrahydroborate

Chemical name	Concentration (%)	Formula	TSCA	EC-No.	CAS RN
Sodium borohydride	≥ 92	NaBH4	Listed	241-004-4	16940-66-2

4. First aid measures

First aid measures

First-aid measures after inhalation : Remove the victim to fresh air, and make him blow his nose and gargle.

First-aid measures after skin contact : Sweep down sodium borohydride that attached on skin with clean, dry cloth and wash thoroughly with water. If necessary, get medical attention.

First-aid measures after eye contact : Gently rinse the affected eyes with clean water for at least 15 minutes and wash with 3% boric acid water solution, Get medical treatment.

First-aid measures after ingestion : Induce the victim vomiting and get medical treatment.

Personal Protection in First Aid and Measures : Rescuers should wear proper protective equipment like rubber gloves, goggles.

5. Fire fighting measures

Suitable extinguishing media : Dried sodium carbonate, sodium chloride graphite powder

Unsuitable extinguishing media : Water, foam, carbon dioxide

Fire hazard : Decomposes by heating or contact with water to generate hydrogen gas that easily ignites.

Firefighting instructions : In case of fire, cover fire area with sodium carbonate, sodium chloride, graphite powder to extinguish.

Personal protection (Emergency : Firefighters should wear protective equipment.

response)

6. Accidental release measures

Personal Precautions, Protective Equipment and Emergency Procedures

General measures : Wear proper protective equipment and avoid contact with skin and inhalation of dust. Conduct operations from upwind and evacuate people downwind.

Environmental precautions

Environmental precautions : Attention should be given to avoid damage to the environment by flowing of spillage to rivers.

Methods and Equipment for Containment and Cleaning up

For containment : Sweep up the chemical and place in a chemical waste container.

7. Handling and storage

Handling

Technical measures : Wear appropriate protective equipment to avoid contact with skin. Avoid contact with water.

Loosen the cap gradually because the decompositon of the chemical cause the increase of inner pressure and the content may fly out from the bottle.

Precautions for safe handling : Avoid formation of dust and aerosols.

Storage

Storage conditions : Store the bottle tightly closed in a cool, dark place because the substance is hygroscopic.

Material used in packaging/containers : Glass, polyethylene, polypropylene.

8. Exposure controls / Personal protection equipment

ACGIH TWA	2 mg/m³ (I) (as B)
ACGIH STEL	6 mg/m³ (I) (as B)

Appropriate engineering controls : Install a local ventilation system in case of dusty condition.

Protective equipment

Respiratory protection : If necessary, wear dust mask

Hand protection : Impervious protective gloves

Eye protection : Safety goggles

Skin and body protection : Protective clothing, protective boots

9. Physical and chemical properties

Physical state : Solid

Color : White

Odor : Odorless

pH : The aqueous solution is alkaline.

Melting point : 36 - 37 ° C

Freezing point : No data available

Boiling point	:	No data available
Flash point	:	No data available
Auto-ignition temperature	:	No data available
Decomposition temperature	:	400 - 500 ° C
Flammability (solid, gas)	:	Flammable solid
Vapor pressure	:	No data available
Relative density	:	1.074
Density	:	No data available
Relative gas density	:	No data available
Solubility	:	Organic solvents: Soluble in ethylenediamine, pyridine. Water: 35.5 % (25℃)
Partition coefficient n-octanol/water (log Pow)	:	No data available
Explosive limits (vol %)	:	No data available
Viscosity, kinematic:	:	No data available
Particle characteristics	:	No data available

10. Stability and reactivity

Reactivity	:	It has a reducing property and reduces aldehydes and ketones.
Chemical stability	:	Stable under normal conditions. Hygroscopic.
Possibility of hazardous reactions	:	Hydrolysis produces sodium tetraborate and flammable hydrogen. Reactions with acids can cause fever and produce toxic diborane.
Conditions to avoid	:	Light, heat, moisture.
Incompatible materials	:	Acids, oxidizing substances.
Hazardous decomposition products	:	Hydrogen.

11. Toxicological information

Acute toxicity (oral)	:	Toxic if swallowed rat LD50=160mg/kg
Acute toxicity (dermal)	:	Classification not possible
Acute toxicity (inhalation)	:	No classification (gas) No classification (vapor) Classification not possible (dust, mist)
Skin corrosion/irritation	:	Causes severe skin burns Although there are reports of "corrosive (cause burn)" for humans, since the details are not clear, the substance was classified into category 1A.
Serious eye damage/irritation	:	Causes serious eye damage Although the details are not clear, there are reports of "corrosive (cause burn)" for humans. Based on a report that when 1 mg was instilled into the eye of rabbits, irreversible damage was produced, the substance was classified into category 1.
Respiratory sensitization	:	Classification not possible
Skin sensitization	:	Classification not possible
Germ cell mutagenicity	:	Classification not possible
Carcinogenicity	:	Classification not possible
Reproductive toxicity	:	Classification not possible

STOT-single exposure	:	May cause respiratory irritation Based on reports as human effects that "may produce irritation of the nasal mucous membranes and the respiratory tract" and "can irritate the nose and throat and the lungs causing coughing and/or shortness of breath", the substance was classified into category 3 (respiratory tract irritation).
STOT-repeated exposure	:	Classification not possible
Aspiration hazard	:	Classification not possible

12. Ecological information

Ecotoxicity

Aquatic acute	:	Classification not possible
Aquatic chronic	:	Classification not possible

Persistence and degradability

No additional information available

Bioaccumulative potential

No additional information available

Mobility in soil

No additional information available

Hazardous to the ozone layer

Ozone	:	Classification not possible

13. Disposal considerations

Ecology - waste materials	:	Add the chemical in diluted sulfuric acid gradually to decompose. Use with local exhaust ventilation because of the generation of hydrogen gas. After decomposition, neutralize the solution and flush in a drain. Or entrust approved waste disposal companies with the disposal.
Contaminated container and packaging	:	In case of disposal of empty bottles, dispose bottles after removing the content thoroughly.

14. Transport information

International Regulations

Transport by sea(IMDG)

UN-No. (IMDG)	:	1426
Proper Shipping Name (IMDG)	:	SODIUM BOROHYDRIDE
Packing group (IMDG)	:	I
Transport hazard class(es) (IMDG)	:	4.3

Air transport(IATA)

UN-No. (IATA)	:	1426
Proper Shipping Name (IATA)	:	Sodium borohydride
Packing group (IATA)	:	I
Transport hazard class(es) (IATA)	:	4.3
Marine pollutant	:	Not applicable
MFAG-No	:	138

15. Regulatory information

Regulatory information with regard to this substance in your country or region should be examined by your own responsibility.

16. Other information

Data sources : Encyclopaedia Chimica, Kyoritsu Shuppan Co, Ltd.(1963).
Handbook of Dangerous Substances Springer-Verlag Tokyo(1991).
Handbook of 17221 Chemical Products, The Chemical Daily Co.(2021).
NITE Chemical Risk Information Platform (NITE-CHRIP), National Institute of Technology and Evaluation.

The information contained herein is based on several references and the present state of our knowledge. However the SDS does not always cover all information about the product, handle the product carefully.The information is intended to ordinary usage, in case of particular handlings, conduct appropriate safety measurements.The information herein is only provision of information, and it does not represent a guarantee the properties of the product.The Safety Data Sheet(SDS) is prepared based on JIS Z7253.